Bishesta Bohara

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Education

The University of Southern Mississippi — Hattiesburg, MS
Bachelor of Science in Computer Science and Mathematics (GPA: 3.762) — *Aug. 2024 – May 2028*

Technical Skills

Programming: Python, JavaScript, TypeScript, Go, C/C++, SQL, HTML/CSS
ML / AI & Data: pandas, NumPy, Matplotlib, scikit-learn, PyTorch, RAG, LLM workflows
Web, Tools & DB: React, Next.js, Flask, FastAPI, Node.js, REST APIs, Git, Docker, PostgreSQL, SQLite
Cloud & Infra: GCP, Vertex AI, Google ADK, MongoDB, Daytona

Experience

Research Assistant & Software Developer — Nov 2025 – Present
Institute for Advanced Analytics and Society — *Hattiesburg, MS*

- Built a NOAA-funded enterprise AI research platform that generates ROI-focused analysis from complex datasets, using RAG pipelines and LLM-driven workflows on a Next.js and Go stack with MongoDB.
- Developed AI agent workflows with Google's Agent Development Kit (ADK) and Vertex AI to support multi-step reasoning and scalable LLM orchestration.
- Architected backend orchestration for containerized Docker sandbox environments on GCP using Daytona, enabling reproducible experimentation and isolated model execution.
- Automated deployments by building CI/CD pipelines, reducing manual release effort and supporting enterprise-scale cloud operations.

Software Engineer Intern — May 2026 – Aug 2026
AI Business Consulting Group (AIBCG) — *Leander, TX*

- Developed AI-powered customer service agents for 3 client businesses (Veteran Realty, Parinama Consulting, and Parinama Academy), automating scheduling, inquiry handling, and repetitive support tasks.
- Reduced manual customer response workload by approximately 30% by designing multi-step agent workflows that retrieve client-specific information and complete routine tasks automatically.
- Deployed a reusable agent framework across all three clients, cutting setup time for each new business use case.

Software Engineer Intern — May 2025 – July 2025
Lavner Education — *Dallas, TX*

- Built full-stack internal applications using React, Node.js, and Python, developing responsive frontend interfaces and backend services to streamline internal operations.
- Designed and implemented REST APIs connecting React dashboards to backend data services for real-time monitoring and reporting.
- Developed automated data pipelines to parse and normalize device telemetry and log data, transforming raw data into visual insights for troubleshooting.

Projects

Clinic Reconciliation Engine | *Python, FastAPI, PostgreSQL, React, scikit-learn, Healthcare AI*

- Built a healthcare reconciliation platform that compares and normalizes clinical records across 3 disconnected EHR systems, using Python, FastAPI, and PostgreSQL.
- Designed ML-assisted workflows for duplicate detection and patient-record matching, achieving 95% matching accuracy on test data.
- Engineered backend pipelines for EHR ingestion, transformation, and validation, improving interoperability and downstream analytics readiness.
- Implemented API-driven search and analytics interfaces supporting automated reconciliation over AI-ready healthcare datasets.

NDR Detection Engine | *Python, pandas, scikit-learn, Zeek, Isolation Forest, ML*

- Built a machine learning-based Network Detection and Response (NDR) engine that analyzes Zeek logs to flag anomalous traffic and beaconing behavior.
- Engineered behavioral features from packet metadata, including interval consistency, byte distribution, and after-hours activity, to drive anomaly detection.
- Trained an Isolation Forest model on network telemetry to detect beaconing, exfiltration, and scanning without hardcoded thresholds, reaching over 90% detection rate at under 5% false-positive rate.
- Developed alert-classification logic that enriches detections with behavioral explanations and severity scoring for faster triage.

Extracurriculars

Organizations: WISE, Grace Hopper, Girls Who Code, Society of Women Engineers (SWE), Rewriting the Code, WiCyS
Honors : President's List (Fall 2024, Spring 2025), Dean's List (Fall 2025, Spring 2026)